

Windows Logging Analysis & Event ID Guide

Security Operations Center (SOC) Technical Lab Report

1. Executive Summary & Overview

Windows Event Logging serves as the primary data foundation for Security Operations Center (SOC) telemetry, threat hunting, and incident response. Windows natively records system, application, and security-related telemetry into standardized logs accessible via the Windows Event Viewer utility. SOC analysts leverage key Event IDs from the Security Log to detect unauthorized access attempts, privilege escalation, lateral movement, and persistence mechanisms across an enterprise network.

2. Core Windows Log Categories

The Windows Event Log architecture categorizes events into three core Windows Logs:

- Security Log:** Tracks audit events such as valid/invalid logons, resource access, privilege usage, and user account modifications. Essential for threat detection.
- System Log:** Records events logged by Windows system components, driver failures, system startup/shutdown, and hardware service disruptions.
- Application Log:** Captures events logged by local applications and software services running on the operating system (e.g., SQL server, custom line-of-business apps).

3. Core SOC Event IDs (Top 5 Essential Controls)

Event ID	Event Name / Category	SOC Significance & Security Meaning
4624	Successful Account Logon	An account successfully logged on. Critical for establishing baseline user behavior and identifying credential usage across network hosts.
4625	Failed Account Logon	An account failed to log on. Indicates potential brute-force attacks, password spraying, or misconfigured automated credentials.

4688	New Process Creation	A new process has been created. Essential for detecting malicious command execution, malware activity, and LOLBins (Living off the Land) execution.
4720	User Account Created	A user account was created. High-priority alert for potential persistence mechanisms established by an adversary after initial compromise.
4672	Special Privileges Assigned	Special privileges assigned to new logon. Triggers when elevated/administrative rights (e.g., SeDebugPrivilege) are granted during a logon session.

4. Practical Investigation: Failed Logon (Event ID 4625) Analysis

To perform detailed forensic investigation, the Windows Event Viewer standard GUI was opened on the target machine. The screenshot below illustrates the captured Security Log displaying an active Audit Failure event (Event ID 4625).

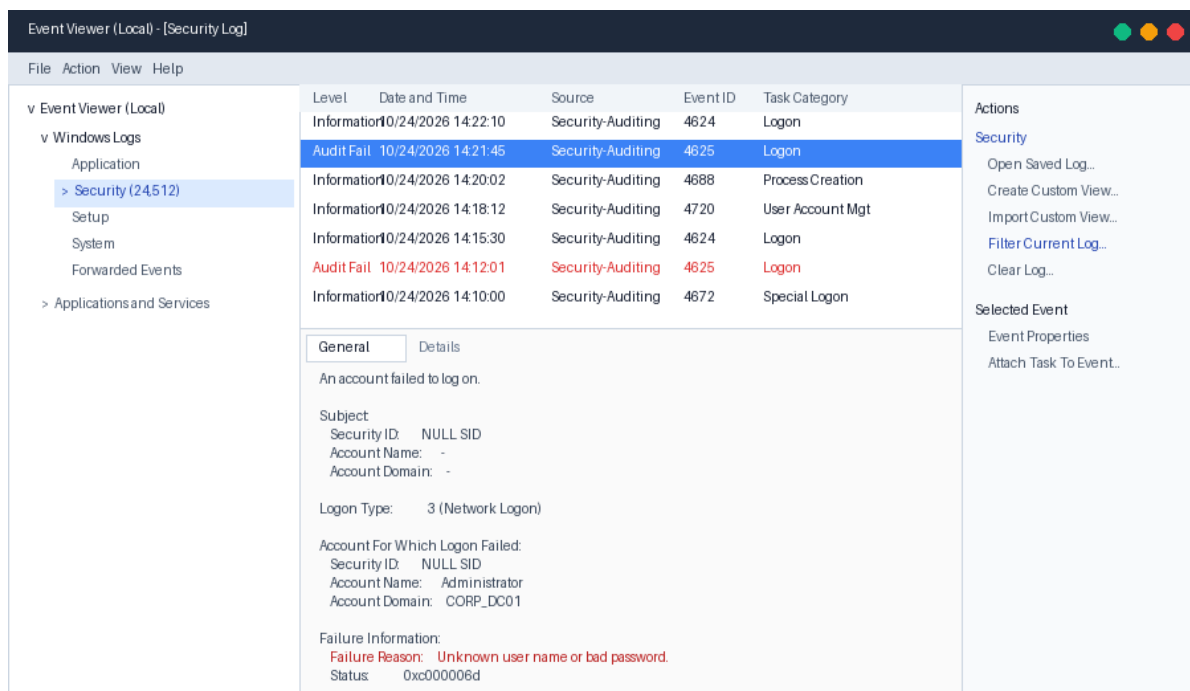


Figure 1: Windows Event Viewer interface showing detailed field breakdown of Security Event ID 4625.

Key Security Fields Breakdown for Event 4625

- **Target Account Name: Administrator** — Identifies the user account targeted by the authentication attempt.

- **Logon Type: 3 (Network Logon)** — Specifies how the user attempted to log on. Type 3 indicates network logon (e.g., SMB share, WinRM, or network resource authentication).
- **Failure Reason: Unknown user name or bad password** — Primary cause of authentication refusal.
- **Status Code: 0xc000006d** — NTSTATUS code indicating the logon attempt failed due to invalid credentials.
- **Sub Status Code: 0xc000006a** — Specific sub-status indicating 'User password mismatch' (wrong password supplied for valid user).
- **Source Network Address: 192.168.1.105** — The IPv4 address of the originating remote host triggering the request. Essential for IP blocking/containment.
- **Workstation Name: ATTACKER-PC** — Host name reported by the source device initiating the logon attempt.

5. SOC Triage & Actionable Recommendations

When detecting high volumes of Event ID 4625 from a singular source network address:

- **Correlate frequency:** Multiple 4625 events within a short threshold (e.g., >10 failures in 1 minute) indicate active automated brute-forcing or password spraying.
- **IP Containment:** Block the source IP (192.168.1.105) at the border firewall / host-based firewall level.
- **Check account status:** Confirm whether targeted accounts (e.g., Administrator) are locked out or compromised via subsequent Event ID 4624 events.





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